Attaching Sensors To A Single Print Optimized Fin-ray Gripper

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***Abstract*—Soft grippers enable safe, adaptable manipulation of objects with uncertain shape. Industrial deployment still faces challenges in manufacturability, reliable feedback, and calibration stability. This review synthesizes research on Fin-Ray gripper mechanics and internal-topology trade-offs, FDM printing of TPU and process–property effects on repeatability, and sensing methods for grasp success, slip indication, and force estimation. Emphasis is placed on practical calibration issues like hysteresis, drift, and unit-to-unit variability, and on approaches that reduce recalibration burden. The paper identifies an integration gap in combining FDM-friendly Fin-Ray optimization with serviceable sensing and calibration-aware feedback for scalable use.**

***Index Terms*—Soft Robotics, 3D Printing, Fin-Ray Gripper, Sensor, Feedback**

I. Introduction

Grippers enable robots to interact with objects inside their workspace and are necessary for executing tasks such as pick-and-place. Gripping can be done as an open-loop task, however, by integrating sensors and feedback, robots are able to behave more like human hands. [1]

Soft gripper development has increased in the last couple of years due to their compliance and adaptability. Their main advantage is their many effective degrees of freedom and ability to grasp objects with different shapes. The main inspiration for soft robots are biological structures and principles. Compared to traditional *hard* robots, soft robots mimic the behaviour of living things, as well as increase safety in Human-Robot Interactions. [2]

Multiple industries are benefited from the development of soft robotic grippers. One of the most prominent is the agriculture industry. This is mainly due to harvesting being very labour intensive, and labour availability decreasing. Additionally, rigid grippers may damage the fruits. [3]

The ultimate goal with soft robotics is to develop an anthropomorphic soft gripper. For task-dependent manipulation, having safe and dexterous grasping is crucial, therefore it is essential to have proprioceptive sensation via embedded flexible sensors. One of the main challenges is the ability to mimic the human hand with controllable stiffness to achieve robust grasping and manipulation. [4]

This review synthesizes Fin-Ray design optimization, FDM TPU manufacturability, and practical sensing/calibration methods to motivate a single-print gripper architecture that enables reliable feedback without multi-material printing.

In agricultural settings, soft grippers have demonstrated an increase in efficiency and productivity for farms. They are able to harvest fruits and vegetables and handle delicate crops, all while minimizing environmental impacts. Numerous industry applications have emerged for soft grippers as well. Growing demands for low-cost automated systems on the food industry have incentivized the development of soft grippers that meet the hygiene restrictions. [2]

With the presence of robots dramatically increasing in human spaces, interaction between humans and machine play an important role. The emerging Industry 5.0 has as a pillar Human-Robot Interaction (HRI) which strongly correlates to soft robotics. Soft mechanisms are flexible and compliant, making them gentle and safer, than their rigid counterparts. They are considered the natural option for HRI due to lower impact forces and skin-friendly interfaces. Another key aspect of the recent growth of soft robotics is the development of material science. Fabrication of soft robots has become more manageable due to developments in manufacturing as well. [5]

II. Fin-ray Grippers

A. Principle

In 1997, Leif Kniese invented the Fin-Ray gripper inspired by the deformation of fish fins. Their main characteristic is that when a load is applied, it does not bend away from it, but rather towards it. This effect is particularly useful to grasp objects by using contact forces to wrap an object and perform simple manipulation tasks. [6]

Fin-Ray grippers are soft robotic devices capable of adapting their shape to conform to a grasped object. They are one of the most researched compliant structures, and can be easily actuated by external devices, independent from the gripper main structure. They have multiple designs that help increase performance based on the object shape and size. [6] The grasping ability of a gripper is directly influenced by the grasping force generated by the actuator and the friction force between the finger and the object. [4]

B. Design Optimization and Customization

Fin-Ray's internal structure controls the adaptability and grasping capabilities for the gripper. The internal beam/rib

topology controls the trade-off between adaptability and contact force. A design without crossbeams maximizes compliance and is best for grasping delicate objects. Adding vertical beams to the cross-beams decreases adaptability, but increases contact force. Finally, a branch configuration enhances final contact force, while improving adaptability, with respect to the classical Festo structure. [6]

Fin-Ray fingers have a specific parametric structure design shown in fig. 1. The height of the finger is denoted as H, while the widths are denoted as W_1, W_2. This forms a trapezoidal shape, which increases grasping ability. Higher stiffness in the back beam allows for a larger percentage of positive contact forces, while simultaneously minimizing lateral deformation. This is why the back beam is usually wider than the front (internal) beam. [6]

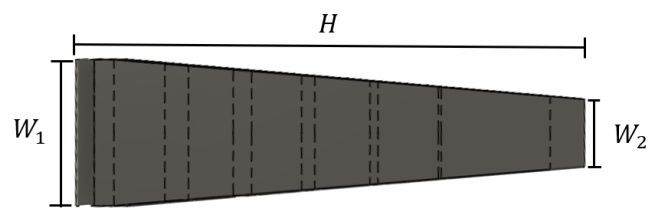

Fig. 1: Festo Fin-Ray Finger Side View (Author's illustration)

Seeing the finger on a frontal view in fig. 2, L represents the width of the base, S the spacing between crossbeams, t the width of the crossbeams, β the angle of the crossbeams, and α the angle between front and back beam. In the case of the branched beam, θ control the angle of the branches. [6]

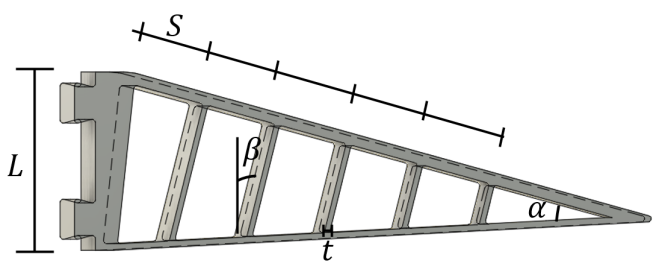

Fig. 2: Festo Fin-Ray Finger Top View (Author's illustration)

C. Known Limitations and Failure Modes

A known limitation for Fin-Ray grippers is that they perform best when contact occurs near the mid-length of the finger. This yields the largest deformation and makes the fingers wrap the object. This means the objects must be raised to have some clearance from the ground or that the grasping pose must be carefully selected. [6]

Jiaqiang Yao et al. expand on some other limitations of certain configurations of fin-ray fingers. When the angle of the crossbeams β is negative, fingertips can close further into the object, benefiting shape adaptation. However, too large of an incline may produce structural interferences. [6]

III. Additive Manufacturing for Soft Grippers

A. 3D Printing

Additive manufacturing enables production of complex geometries and rapid prototyping. By manufacturing layer by layer, functional end-use components can be formed. [7] Fused Deposition Modelling (FDM) is a 3D printing technique that allows the construction of soft robotic structures. Georgopoulou et al. devised a method using multi-material FDM to create TPU structures with embedded piezoresistive elements that act as strain sensors. [8]

The main printing process parameters include feed rate, infill density, pattern and geometry, build orientation, layer height, print speed, wall thickness, and extrusion nozzle and bed plate temperature. [9]

Increasing the infill density leads to an increase in Young's modulus of elasticity, yield, and ultimate strengths. Additionally, regarding build orientation, specimens printed flat or on-edge show significantly higher ultimate stress compared to upright printed specimens. Finally, an increase on wall thickness and count results in increased percent elongation at failure, tensile strength, Young's modulus, and toughness. [9]

A conventional soft or rigid gripper may be comprised of multiple pieces and require an additional assembly step. Yang et al. experimented with designing a gripper that has the whole body 3D printed. This reduces the need for assembly and makes the gripper light weight and soft. [10]

Their gripper consists of a hollow structure with an upper plate that connect the three soft fingers. This way all fingers bend with a single actuator by pulling a wire from the middle of the plate, instead of having an individual one for each finger. This reduces cost, complexity, weight and size of the gripper. With this model, a relationship between the angular displacement of the finger tip and the length of the pulling wire movement can be investigated to characterize the gripper. [10]

B. TPU

Thermoplastic polyurethanes (TPUs) are suited for fused deposition modelling of pieces that require flexibility and strength, like a gripper. The behaviour of the TPU, in terms of deformation is highly dependant on the printing process parameters. Therefore predicting large deformations may pose a challenge. [9]

TPU has high ductility, abrasion and hydrolysis resistance. The properties of the material are directly influenced by parameters such as bed temperature, nozzle temperature, and build orientation [7]. By combining it with carbon black fillers, conductive filaments can be created that are compatible with commercial 3D printers. [8]

TPU's melting temperature ranges from 120°C to 200°C, but vary by formulation. It begins to disintegrate at 250°C, in which polymer chains start to break down and the material loses its mechanical characteristics. [11] In Georgopoulou et al.'s setup for manufacturing a soft gripper with TPU, it is recommended to use 230°C for the nozzle temperature. Below 230°C there is under-extrusion, and above 230°C excessive

stringing. Setting the bed temperature to less than 45°C may cause insufficient adhesion to the bed plate. [8]

IV. Sensing for Soft Robotics

A. Grasp Success and Slip Detection

A slip event can be defined as when two surfaces slide against each other, generating friction in the opposite direction of motion. Slipping may lead to the partial or even the complete loss of the object being grasped. These events occur when there is an imbalance of the normal and shear forces between the object and the fingers. Additionally, some tasks involve additional external forces that may disrupt this equilibrium and the object's stability. [12]

In fields like logistics, robots face challenges in complex assembly lines when dealing with objects with unknown physical characteristics. They struggle to detect and correct grasping instability, as many algorithms run as open-loops. Slip detection is crucial for adjusting force and for texture discrimination. [12]

An ideal embedded sensor must be compliant to the morphology of soft robots. They should have minimum effects on the dynamics of the robot's actuation. [6]

Development of slip detection sensors for robotics has been a challenge. Hossain et al. proposed a multi-axis stretchable sensor with integrated carbon nanotubes to address this challenge. The sensor is able to detect both normal and shear forces based on deformation-induced changes on resistance. It can detect slip events with varying speed, levels and surfaces. Applications for this type of sensors range from automated manufacturing, to service robotics. [12]

On the other hand, piezoresistive sensors can be used to detect obstacles preventing the gripper from opening. These components are successful at detecting abnormal grasp events (e.g., blocked motion/obstacles) in real time. In cyclic tests, a blockage during opening produced a distinct sudden resistance drop, and the signal returned to the baseline once the obstacle was removed, after which normal cycling resumed. [8]

B. Force Estimation

Piezoresistive sensors embedded into soft actuators can be useful for force estimation or strain sensing. Georgopoulou et al. sensors exhibited good linearity up to 30% strain, enough to cover the deformation range of the gripper. They were able to differentiate between big and small objects, based on the readings. Embedded strain sensors can discriminate grasp conditions (e.g., object size) because different contact geometries produce different finger deformation patterns, which map to different resistance changes. [8]

It is possible and viable to attach an off-the-shelf pressure sensor on the surface of a gripper finger in order to monitor contact forces. The sensor itself does not need to be manufactured with the gripper using complex techniques. Yang et al. work proves this. [10]

In Yang et al. work, the relationship between the force feedback and the displacement of a pulling wire to actuate the gripper was investigated. The curve produced by this relationship changed between object of varying sizes and shapes. Force exerted by the gripper on an object increases linearly at the beginning of the gripping process. When force reaches a specific value it slows down. This value is dependent on the mass and size of the object. The larger the mass and the size, the larger the specific value is. [10]

Due to the differences in the force curves, it is possible to estimate what object the gripper is grasping. This detection can be done based purely on the force-feedback information. [10]

C. Calibration, Drift and Uncertainty

Strain sensors show different resistance values for same deformations depending on whether the actuator is loading or unloading. The maximum difference between the loading and unloading curves is denoted as hysteresis (H). Even if hysteresis remains constant, the baseline resistance is constantly shifting, reducing long-term accuracy. This phenomenon is known as drift. This is shown in fig. 3. [13]

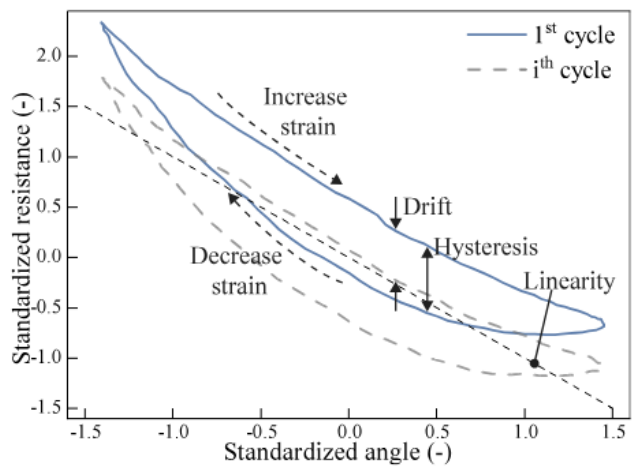

Fig. 3: Hysteresis and Drift, Reproduced from Ji et al., "Design and calibration of 3d printed soft deformation sensors for soft actuator control", 2023, Fig. 4

A known approach for practical calibration is to approximate the resistance-deformation relationship with a linear model using standardized measurements. With this approach, calibration quality can be calculated using an average RMSE over all cycles. [13] Since soft sensors show large nonlinearities and hysteresis, sometimes it is very difficult to use these linear approaches. [14]

On the other hand, DongWook et al. proposes an adaptive calibration method for soft sensors that captures the change in characteristics through domain adaptation, using optimal transportation. An offline calibration is performed first for alleviating difficulties. Then, an online calibration is performed to compensate for the parameter changes produced by continuous use over an extended period of time. [14]

Deep learning algorithms can be used to map the curve of the soft sensor which includes the hysteresis behaviour. These

algorithms use learning to fit the curve instead of trying to remove it. By using domain adaptation, a time-delay artificial neural network is able to predict drift, essentially removing the need for periodic recalibration and the behavioural differences between sensors. [14]

V. Motivation

A. Identified Innovation Gap

From all the literature review, numerous projects attempt to tackle similar problems using several different approaches. Georgopoulou et al. sensors require access to multi-material 3D printing. This makes them slightly more complicated to manufacture, since you need dedicated equipment. [8]

Hossain et al. sensors use multiple materials, assembled by layers. There are 3 types of layers, insulating layers, force-sensitive layers, and electrode layers. The sensor's electrodes consist of multi-walled carbon nanotubes (MWCNTs) embedded in a polymer matrix. This also involves complicated materials and manufacturing processes. [12]

DongWook et al. method for calibrating soft sensors using an AI model is complex. It requires compiling an extensive dataset and having dedicated hardware to train the model. Although it might be complex, it is particularly useful for mitigating the effects of hysteresis and drift. [14]

Finally, Yang et al. gripper uses resin printing to manufacture a three-fingered soft gripper. The fin-ray design is not optimized, and the sensors do not go any calibration step. Although resin printing is also a type of 3D printing, FDM printers are more widely used worldwide and accessible. [10]

A gap exists in FDM-friendly soft grippers where the geometry is optimized for the Fin-Ray trade-off, sensing is achievable with replaceable off-the-shelf sensors, and calibration methods explicitly address hysteresis/drift without requiring multi-material fabrication or large training pipelines. The goal is to target the co-design of printable Fin-Ray geometry plus sensor mounting architecture and a practical calibration workflow suitable for low-cost replication.

B. Project Focus

The focus of this project is to develop a soft gripper capable of grasping objects with varying sizes, weights, and shapes. It should be easily manufacturable using the widely commercially available FDM technology and TPU material, with minimal supports and no multi-material. This ensures reproducibility without the need of complex machines or manufacturing processes. Finally, the print should require minimal post-processing.

After having an easy-to-print gripper, the goal is to attach off-the-shelf sensing components to measure forces and grasping levels. This will be specially useful in industrial settings that are highly automated. Additionally, by making the sensors a separate entity, they are easily replaceable in case of malfunction, wear or tear. This increases serviceability.

By implementing calibration methods specific for soft sensors to handle hysteresis and drift, the system obtains higher quality measurements and increases reliability. Sensor redundancy also helps mitigate errors in measurement. The feedback outputs should allow the user to know grasp success, confidence, and force/contact levels.

Finally, the gripper must be low-cost to build, have minimum assembly steps and robust wiring. This makes it a high quality product capable of operating in industrial settings or facilitates mass production. To achieve this, single-printing the gripper helps reduce the amount of pieces, assembly steps, weight and size.

VI. Project Development

A. Design

To make the gripper single-print a new design was developed with two fin-ray fingers attached to a base. This is attached to a second platform through lateral supports and diagonal central supports. When a pulling force is applied to the midpoint of the base, a torque is induced inwards, which makes the fingers close and grasp an object. To create this pulling force a wire and servo motor were used, but this design may also be compatible with other actuators, like a piston.

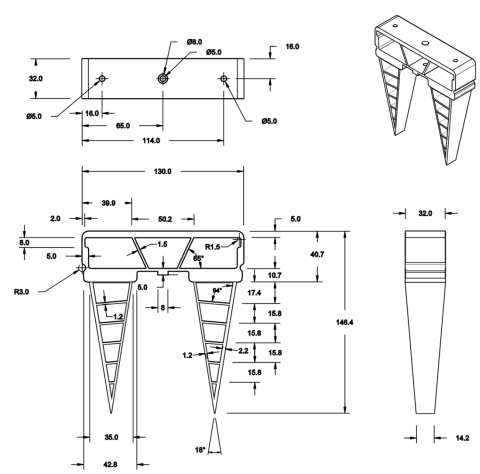

Fig. 4: Technical Drawing

Several finger configurations were tested and evaluated. The configurations are one festo, one branch, one hollow and one with vertical beams. For testing purposes a variation of the gripper was created that allows for swappable fingers, enabling all configurations to be evaluated under identical conditions

The proposed gripper was manufactured using TPU on a commercial 3D-printer. Nozzle temperature was set to 230°C, bed temperature to 50°C, printing speed to 100 mm/s, and infill to 15%. The gripper was printed on a horizontal position to eliminate support structures. An additional structure made from PLA was printed which connected to the gripper to hold the servo-motor. In industrial applications, this structure is replaced with the end-effector of a manipulator for example. A flexible pressure sensor ZD10-100 was attached on the inside part of the fingers using electrical tape that returns an analogue value when in contact with an object.

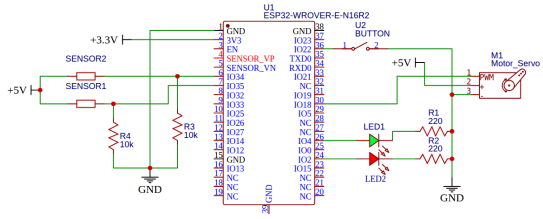

Fig. 5: Electrical Diagram

A pseudo-control strategy (fig. 6) was implemented for grasping. If the sensor value reached a certain threshold, the state of the object was defined as grasped. However, the gripper keeps closing a bit more until it reaches a second threshold to ensure a solid contact. A safety maximum servo angle was defined to avoid damages. For minimizing slipping, a strategy was implemented where once the object is grasped, if contact (pressure) decreases, it starts to increase the motor angle to compensate. This is done until the value reaches the stopping threshold again. A MG996R Servo Motor was used.

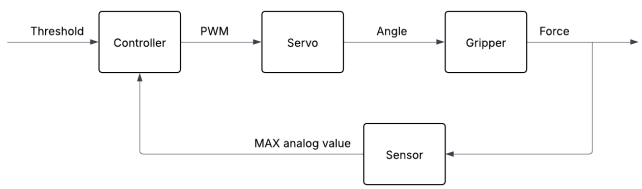

Fig. 6: Pseudo-Control Diagram

Since slight variations exist in sensor readings due to object geometry, contact surface area, and noise, the maximum value between the two sensors was used as the input to the controller. The contact threshold was set to 1200, which is almost 30% of the ESP32's DAC (12-bit). The final closing threshold was set to 3000, which is almost 70% of the DAC. Sensor sensitivity is controlled by the resistor in the voltage divider. Therefore, threshold values were selected through preliminary trials.

B. Testing

A testing rig (fig.7) was used to collect data from the grasping events. Three objects were used to test the grasp capabilities: a sphere, a cylinder and an icosahedron.

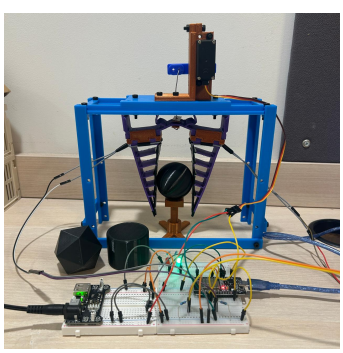

Fig. 7: Testing Rig

The shallow finger was the most compliant; however it lacked sufficient contact force to hold the object steadily. The vertical beam finger was more rigid but lacked compliance to grab uneven surfaces. The branch configuration was in between the previous two; however it would have worked better if the motor had more torque. Therefore, the selected finger was the Festo configuration.

C. Results

The design was able to reduce the number of individual pieces that compose the gripper (excluding the support base), from seven to one. It reduced complexity and the need for assembly. It can be printed on any commercial 3D printer as a single piece with no supports.

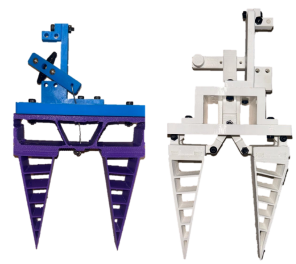

Fig. 8: Single Print vs Multiple Pieces

10 trials of gripping each geometry were done. Angle, and analog values were extracted at each timestep and plotted.

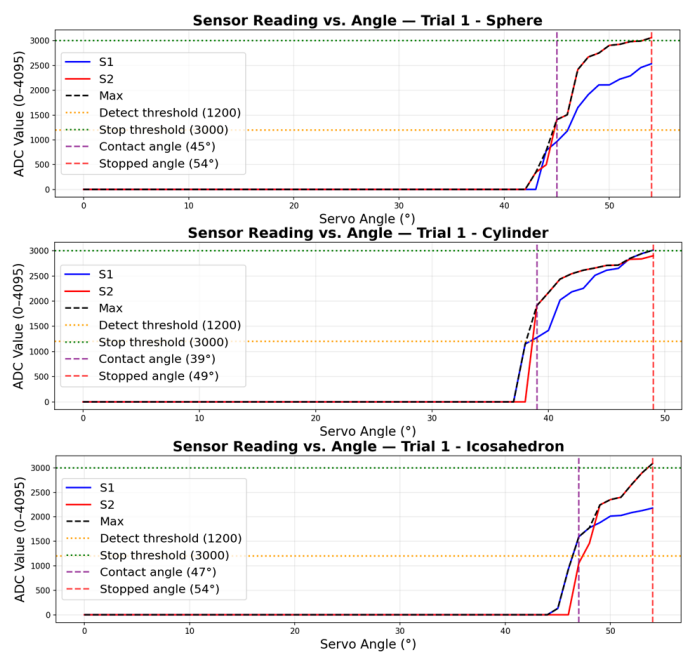

Fig. 9: Grasping Plots

Fig. 9 shows one grasping trial for each geometry. The sphere was detected when motor reached 45°, and it reached the firm grasp threshold at 54°. Its curve resembles approximately linear growth, with noise. The cylinder reached the first

threshold at 39°, and the final threshold at 49°. Its curve rises quickly then slowly plateaus. Finally, the icosahedron met the first threshold at 47°, and the final threshold at 54°. Its curve exhibited the highest variability. This is due to the geometry, as the gripper comes in contact with the object, it first touches the pointy parts, which triggered initial sensor response, but as the object was being grasped, it shifted and flat surfaces came into contact, changing the values.

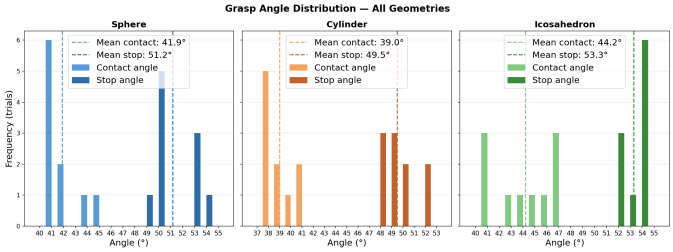

Fig. 10: Motor Angle Frequency Chart

Fig. 10, shows a frequency chart of motor angles when thresholds were met for each geometry across 10 trials. Mean contact angle when grasping the sphere was $41.9° \pm 1.45°$ and the stopping angle was $51.2° \pm 1.81°$, with the most common value being 41°and 50°respectively. For the cylinder, the mean contact angle was $39° \pm 1.24°$ and stopping angle $49.5° \pm 1.5°$, and the most frequent angles were 38°and 48°-49°respectively. Finally, the icosahedron had a mean contact angle of $44.2° \pm 2.57°$ and stopping angle of $53.3° \pm 0.94°$, with the most frequent values being 40°, 47°and 54°respectively.

The cylinder had the lowest mean contact angle, as well as mean stop angle. The sphere had both the middle mean contact angle and the middle mean stop angle. The icosahedron had the highest value for both. These three geometries are very different in terms of how the gripper makes contact and the area available, however, by using a soft material, the fingers were able to adapt and grasp the object regardless of the shape. The cylinder also had the least variation in angles. While the icosahedron appears to have the lowest standard deviation for closing angle, that is only because most of the time measurements did not reach the threshold and the control system stopped at the max angle (54°).

Grasping detection was not perfect for all geometries. The sphere had on average the highest detection rate (95%), with cylinder being second (90%), and icosahedron last (80%). Main source of error came from the object not making sufficient contact with the surface of the sensor.

VII. Conclusion

The Fin-Ray gripper was successful at grasping the sphere, cylinder, and icosahedron using the pseudo-control strategy due to its compliant nature. The implementation of off-the-shelf sensors allowed for an easier manufacturing process, without compromising functionality. The proposed gripper design successfully achieved its primary objective of being single-pieced, printable with any commercial 3D printer using standard TPU material. Additionally, it required no support structures, which reduced time and material.

By using the analog values from the sensors, grasping curves were obtained, which were distinct for each object geometry. Moderate repeatability was observed across trials, however measurement quality can be improved by implementing a filtering system or a calibration method. This would enhance the overall quality of the control system. Actual object grasping was effective almost 100% of the time, just the detection varied across geometries.

A. Future Work

Since the shape and size of an object influences the grasping curve, a future improvement is to train a neural network to identify the size and shape of an object the gripper is grasping. This could help with classification without the need of computer vision or other methods. Sensor fusion could also be explored to improve measurement quality. Additionally, a comparison against a multi-material gripper with embedded sensors would provide a useful performance benchmark.

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